

Localization and Object Detection

A fixed dense tensor becomes a scored set of boxes

Prof. Nipun Batra

ES 667 · Deep Learning · IIT Gandhinagar

**L8B kept spatial features;
detection must spend them**

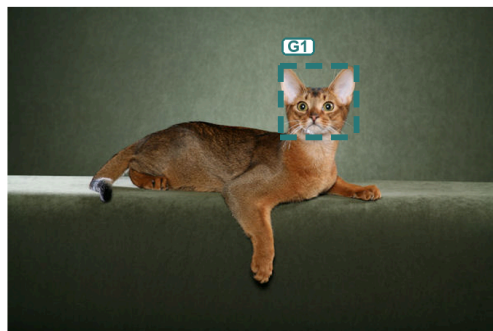
Two real images turn “where” into supervised geometry

A REAL TWO-IMAGE MINI-BATCH

Two photographs · two official XML head boxes

I1 · Abyssinian_1

OBSERVED PHOTO + XML GT



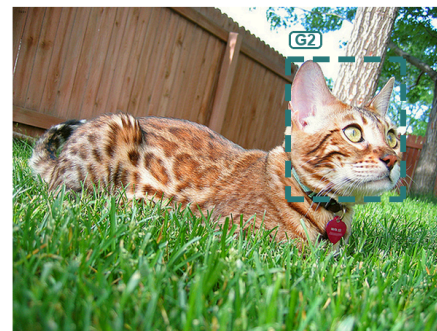
VOC XML (333, 72)–(425, 158)

course **xyxy** [332, 71, 425, 158]

93 × 87 source pixels · ground truth, not model output

I2 · Bengal_10

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VOC XML (315, 60)–(446, 219)

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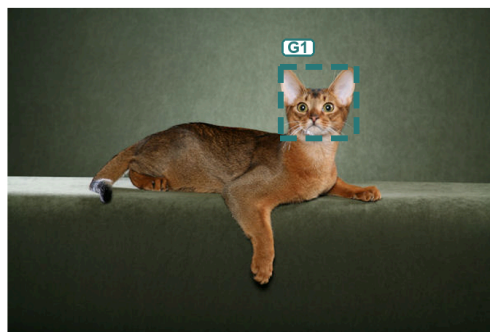
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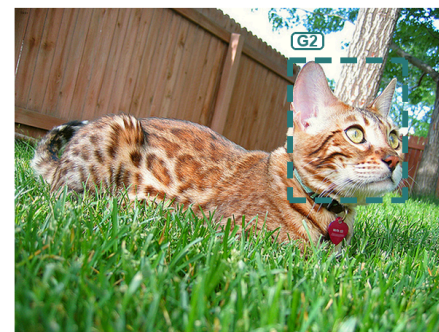
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OBSERVED pixels + official XML head boxes. Detection reuses the L8B backbone, but changes the head, targets, geometry, and evaluation.

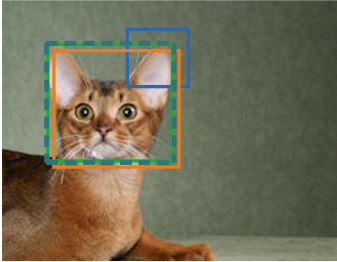
Commit: which operations turn five candidates into trustworthy detections?

QUESTION

HELD CASE · OPENING MINI-BATCH CONSTRUCTED, not model output: A/B/E belong to I1; C/D belong to I2.

THE SAME A-E SPINE, NOW ON TWO REAL PHOTOS
Candidates/scores are constructed; XML boxes are observed

I1 · Abyssinian_1
G1 + A/B/E



A 0.95 B 0.90 E 0.89

I2 · Bengal_10
G2 + C/D



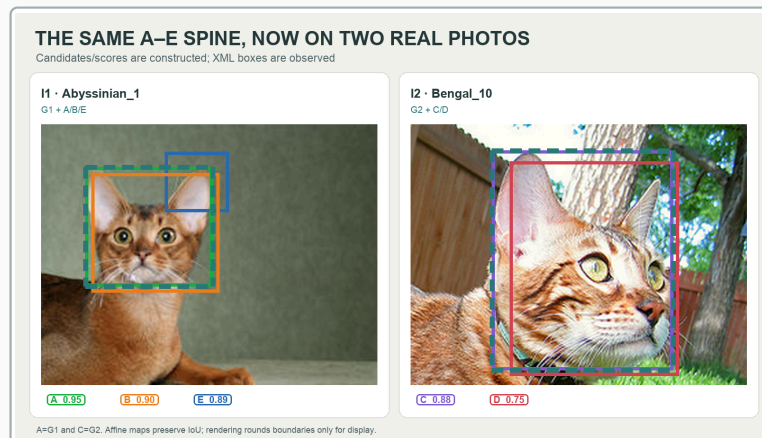
C 0.88 D 0.75

A=G1 and C=G2. Affine maps preserve IoU; rendering rounds boundaries only for display.

Commit: which operations turn five candidates into trustworthy detections?

QUESTION

HELD CASE · OPENING MINI-BATCH CONSTRUCTED, not model output: A/B/E belong to I1; C/D belong to I2.



A · THRESHOLD

score threshold alone

B · GEOMETRY

decode → score → NMS
→ match

C · LOSS

rerun loss at inference

The 80-minute route has one cumulative spine

By the end, you should be able to turn a dense output tensor into trained candidates, filtered detections, and a defensible AP number.

CORE · 55 MIN

box contract → IoU →
dense head →
assignment + loss → NMS
→ matching + AP →
revisit

SHOULD · 15 MIN

Smooth L_1 /GIoU → two-
stage vs one-stage →
FPN → COCO protocol

OPTIONAL · 10 MIN

anchor-free heads →
Soft-NMS / set prediction
→ primary provenance

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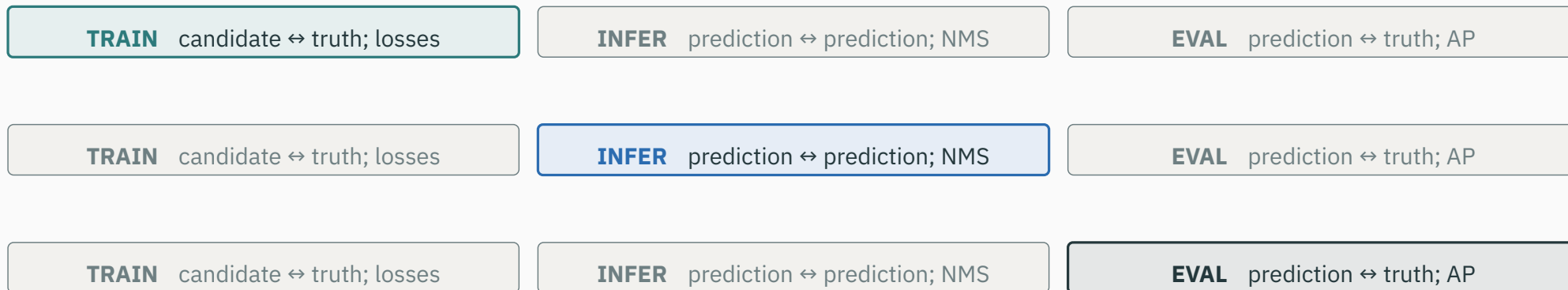
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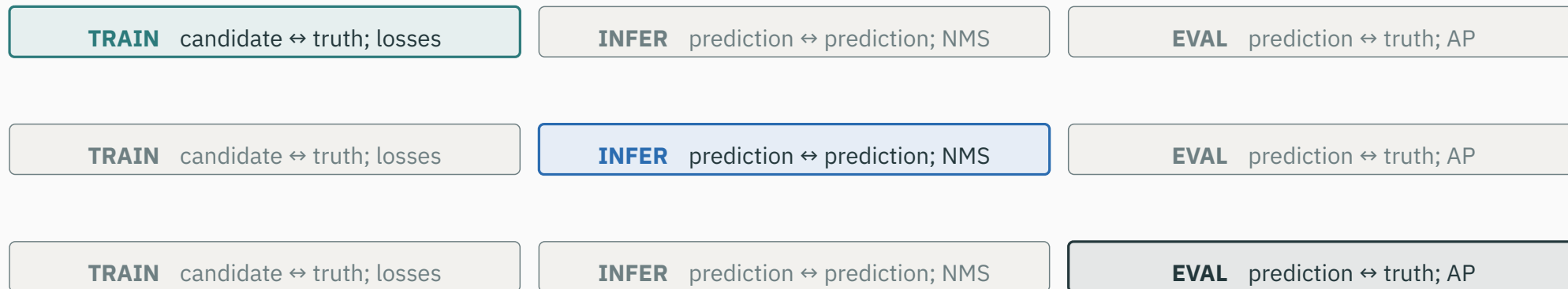
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— given / eval — truth / train — raw candidate — infer / control — survivor / TP — suppressed / FP

Three clocks reuse boxes—but never the same comparison



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TRAIN

assign candidate ↔ truth

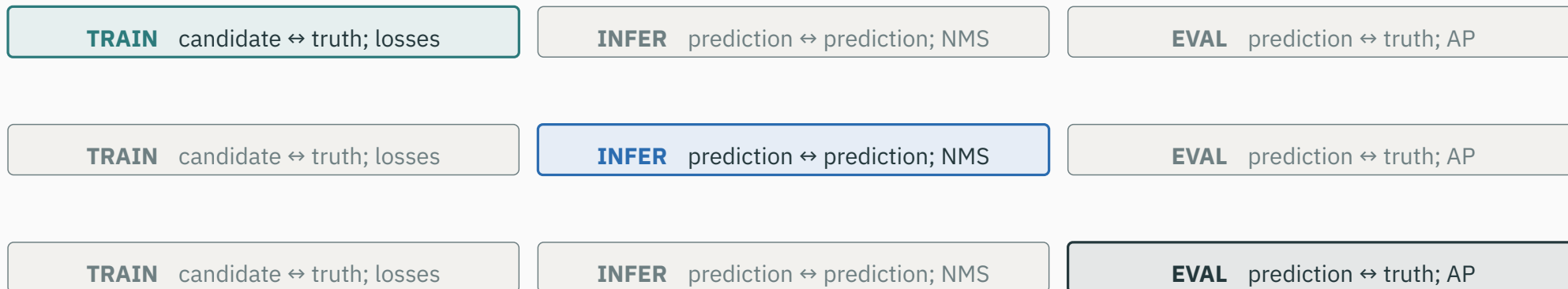
INFER

compare prediction ↔ prediction

EVAL

match prediction ↔ truth

Three clocks reuse boxes—but never the same comparison



TRAIN

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The clock strip will remain visible whenever “IoU threshold” could otherwise mean three different things.

**A box is four numbers with an
explicit convention**

Detection predicts a set, not a longer class vector

classifier: $x \rightarrow \hat{p} \in \Delta^{K-1}$

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detector: $x \rightarrow \left\{ \left(\hat{y}_i, \hat{b}_i, s_i \right) \right\}_{i=1}^N$

CLASS

$\hat{y}_i$: what is here?

BOX

$\hat{b}_i \in \mathbb{R}^4$: where is it?

SCORE

s_i : how confident?

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A fixed network tensor may contain many candidate triples; thresholding and NMS make the returned set variable-length.

Our coordinate convention prevents the silent “+1” bug

D DERIVATION

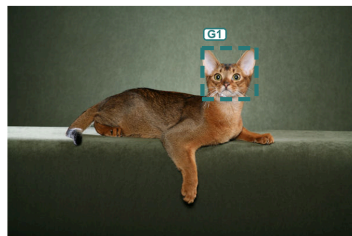
HELD CASE · REAL ANNOTATION CONTRACT VOC XML uses one-based inclusive corners; the course converts once to zero-based half-open $[x_0, y_0, x_1, y_1)$.

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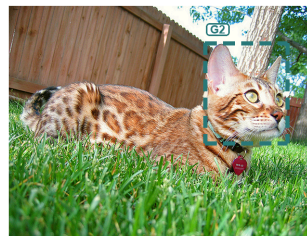
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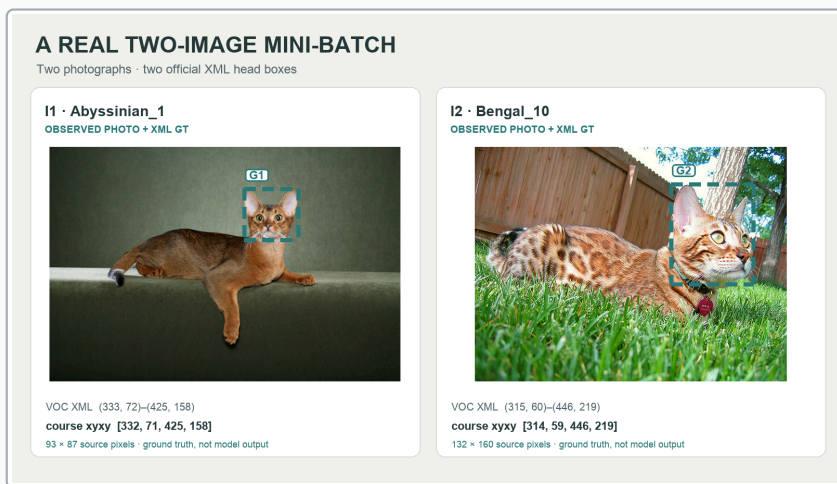
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Example I1: VOC (333, 72)–(425, 158) becomes [332, 71, 425, 158), so area is $93 \cdot 87 = 8,091$. No later component adds +1.

The canonical truths fix every later area calculation

HELD CASE · CONSTRUCTED NUMERIC SPINE $G_1 = [10, 10, 50, 50]$ in I1; $G_2 = [60, 15, 90, 55]$ in I2. Separate positive affine maps preserve IoU.

G_1

$$w = 50 - 10 = 40$$

$$h = 50 - 10 = 40$$

$$|G_1| = 1,600$$

G_2

$$w = 90 - 60 = 30$$

$$h = 55 - 15 = 40$$

$$|G_2| = 1,200$$

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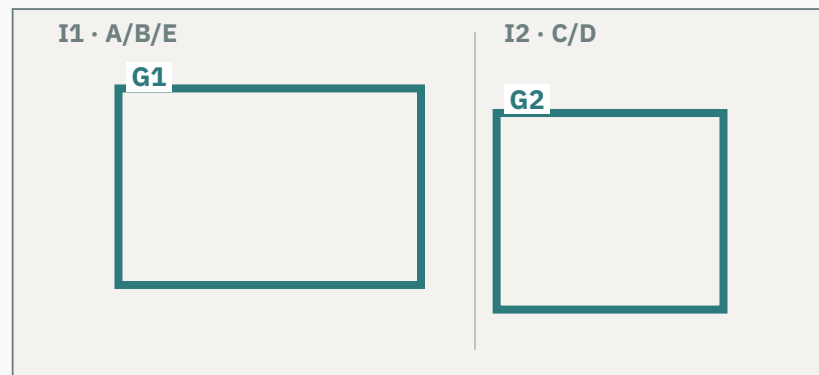
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A localization head shares evidence, then asks two questions

SHARED LOCAL FEATURE

$$h = f_{\theta(x)}$$

→

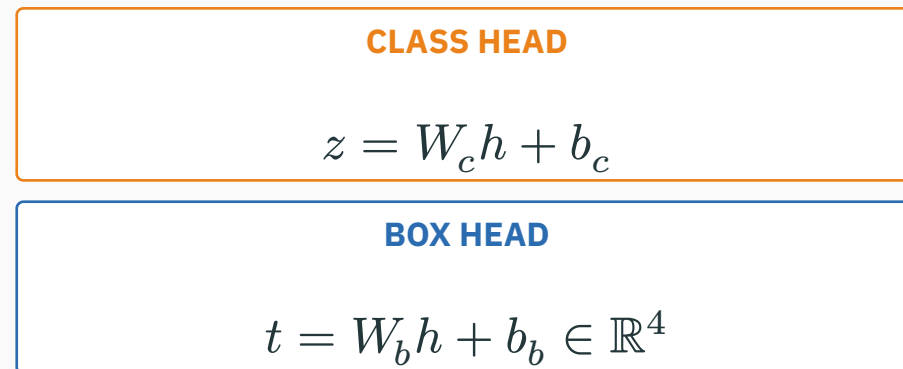
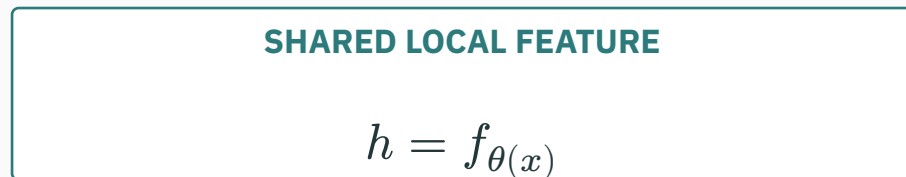
CLASS HEAD

$$z = W_c h + b_c$$

BOX HEAD

$$t = W_b h + b_b \in \mathbb{R}^4$$

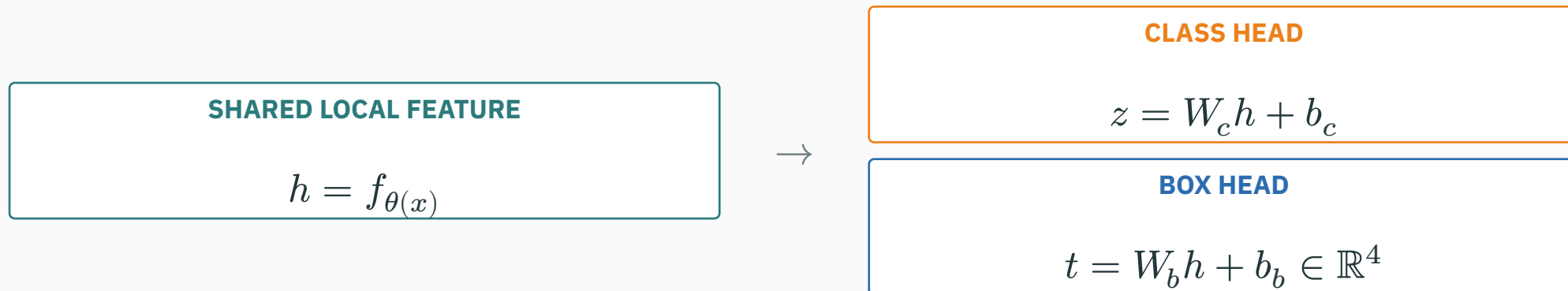
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what is here?

how should the reference move?

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Detection is multi-task learning repeated at spatial locations—not a classifier with four unexplained extra outputs.

Commit: what offsets move this anchor to its matched truth?

QUESTION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

HELD CASE · POSITIVE ANCHOR Reference $a = (50, 50, 40, 40)$; matched target $b = (58, 46, 80, 40)$, both in center-size form.

TRANSLATE

$$t_x = \frac{x - x_a}{w_a}$$
$$t_y = \frac{y - y_a}{h_a}$$

RESIZE

$$t_w = \log\left(\frac{w}{w_a}\right)$$
$$t_h = \log\left(\frac{h}{h_a}\right)$$

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Calculate all four before the answer. Which offset must be zero? Which must be $\log 2$?

The anchor target is a scaled correction, not an absolute box

A ANSWER

TRAIN candidate $\leftrightarrow$ truth; losses

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$$t_x = \frac{58-50}{40} = 0.20$$
$$t_y = \frac{46-50}{40} = -0.10$$

$$t_w = \log\left(\frac{80}{40}\right) = \log 2 \approx 0.693$$
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Translations are measured in anchor sizes; log-scale changes keep decoded widths positive and make zero mean “unchanged.”

Decoding must recover the same geometry

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

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$$\hat{x} = x_a + t_x w_a = 50 + 0.20(40) = 58$$

$$\hat{y} = y_a + t_y h_a = 50 - 0.10(40) = 46$$

$$\hat{w} = w_a \exp(t_w) = 40 \exp(0.693) \approx 80$$

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Encode and decode are inverse contracts; a unit test should recover b before any model is trained.

A YOLO-style anchor head emits nine scalars per candidate

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

HELD CASE · HEAD CONVENTION For this teaching ledger only: one objectness logit + $K = 4$ class logits + four box offsets.

OBJECTNESS

one logit
is anything here?

CLASSES

$K = 4$ logits
which class?

OFFSETS

four numbers
how should the anchor
move?

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$$1 + K + 4 = 1 + 4 + 4 = 9$$

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Teaching convention only: other heads may fold background into classes or predict box sides.

The dense tensor fixes capacity before the object count is known

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

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EVAL prediction $\leftrightarrow$ truth; AP

HELD CASE · TENSOR LEDGER 20×20 locations, $A = 3$ anchors per location, $K = 4$ classes, nine outputs per anchor.

quantity	calculation	count
candidate boxes	$20 \cdot 20 \cdot 3$	1,200
output scalars	$20 \cdot 20 \cdot 3 \cdot 9$	10,800

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The head always emits 10,800 scalars. Labels and post-processing decide which candidate records matter.

Toy assignment says which candidates receive which losses

TRAIN candidate $\leftrightarrow$ truth; losses**INFER** prediction $\leftrightarrow$ prediction; NMS**EVAL** prediction $\leftrightarrow$ truth; AP

For one truth, three anchors have IoUs 0.76, 0.48, and 0.12. Use the declared **toy** policy $\tau_+ = 0.70$, $\tau_- = 0.30$.

IoU	assignment	loss terms
0.76	positive	objectness + class + box
0.48	ignore	no gradient in this toy policy
0.12	background	objectness only

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Assignment is a training rule, not AP matching and not NMS. Real detectors use architecture-specific thresholds or dynamic matchers.

One positive candidate produces a fully auditable loss

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

Target $t = (.20, -.10, .693, 0)$; prediction $\hat{t} = (.25, -.05, .60, .05)$. Let $p_{\text{obj}} = .90$, $p_{\text{cup}} = .80$ and $\lambda = 2$.

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$$-\log .90 \approx .105$$

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OBJECTNESS

CLASS

$$-\log .90 \approx .105$$

$$-\log .80 \approx .223$$

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$$-\log .90 \approx .105$$

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BOX L_1

$$|.05| + |.05| + |-.093| + |.05| = .243$$

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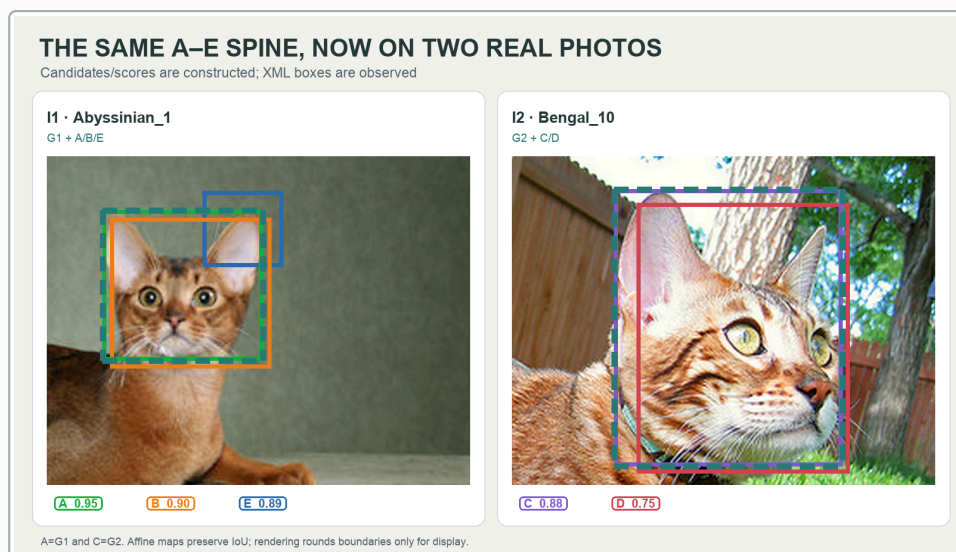
$$\mathcal{L} = .105 + .223 + 2(.243) \approx .815$$

Write the target, parameterization, reduction, and λ . “Box loss” without those contracts is not reproducible.

**IoU makes box quality
geometric**

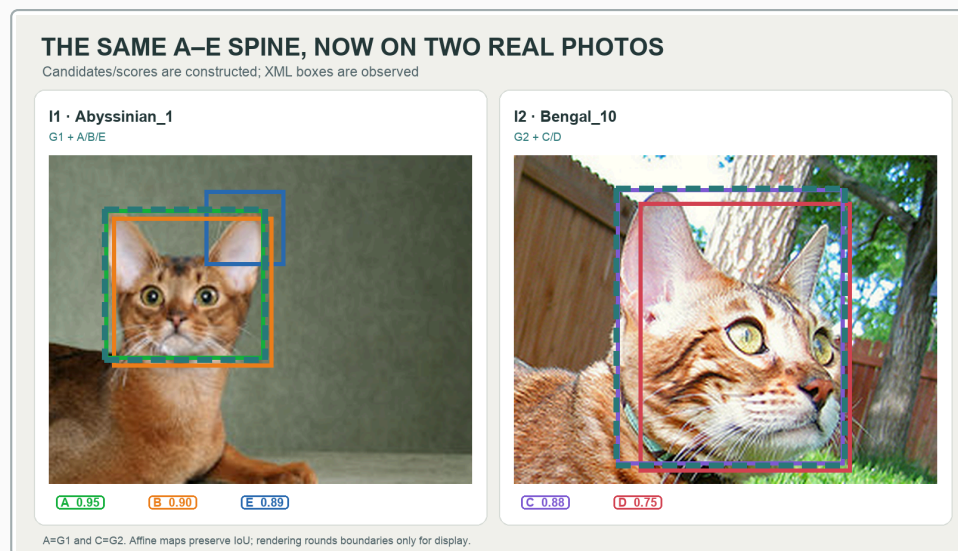
IoU asks how much two regions agree

HELD CASE · SAME I1 PAIR $A = G_1 = [10, 10, 50, 50]$ and $B = [12, 12, 52, 52]$ are affinely placed on the real Abyssinian head ROI.



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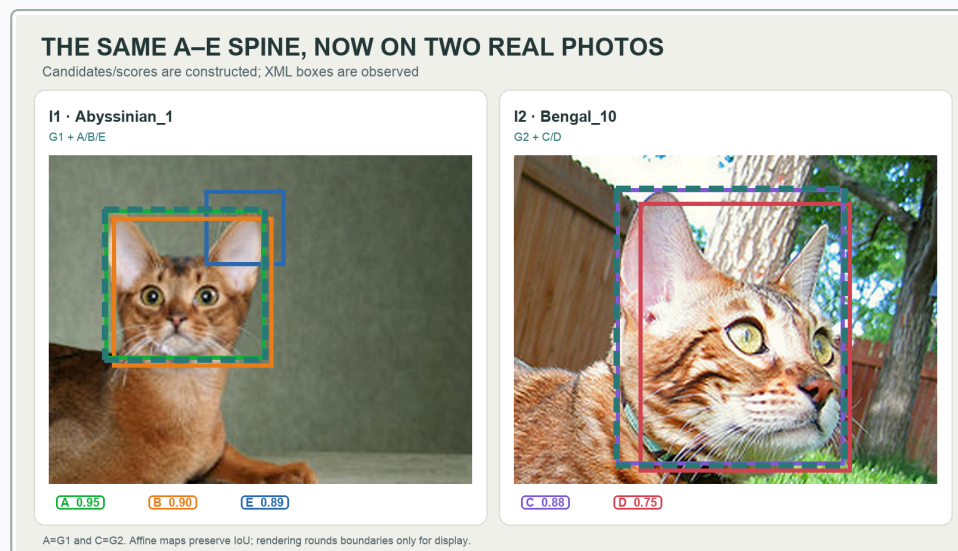
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$$\text{IoU}(A, B) = \frac{|A \cap B|}{|A \cup B|} \in [0, 1]$$

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0: disjoint 1: identical invariant to scaling the whole scene

The held overlap is 0.822, not a visual guess

D DERIVATION

HELD CASE · SAME PAIR $A = [10, 10, 50, 50]$, $B = [12, 12, 52, 52]$; both have area 1,600.

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HELD CASE · SAME PAIR $A = [10, 10, 50, 50]$, $B = [12, 12, 52, 52]$; both have area 1,600.

$$w_I = \min(50, 52) - \max(10, 12) = 38$$

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$$A_I = 38 \cdot 38 = 1,444$$

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$$A_U = 1,600 + 1,600 - 1,444 = 1,756$$

The held overlap is 0.822, not a visual guess

HELD CASE · SAME PAIR $A = [10, 10, 50, 50]$, $B = [12, 12, 52, 52]$; both have area 1,600.

$$w_I = \min(50, 52) - \max(10, 12) = 38$$

$$A_I = 38 \cdot 38 = 1,444$$

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$$A_U = 1,600 + 1,600 - 1,444 = 1,756$$

$$\text{IoU}(A, B) = \frac{1444}{1756} \approx 0.822$$

The same 0.822 will later justify suppressing B around kept box A.

Safe IoU clamps disjoint intersections to zero

D DERIVATION

For each axis, $w_I = \max(0, \min(x_1^A, x_1^B) - \max(x_0^A, x_0^B))$ $h_I = \max(0, \min(y_1^A, y_1^B) - \max(y_0^A, y_0^B))$

Then, $A_I = w_I h_I$ $A_U = A_A + A_B - A_I$
 $\text{IoU} = \frac{A_I}{A_U}$

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Without $\max(0, \cdot)$, two disjoint boxes can produce two negative side lengths—and a meaningless positive “intersection.”

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 $\text{IoU} = \frac{A_I}{A_U}$

Without $\max(0, \cdot)$, two disjoint boxes can produce two negative side lengths—and a meaningless positive “intersection.”

Also assert valid boxes: $x_1 \geq x_0$, $y_1 \geq y_0$, and define the zero-union case explicitly.

SHOULD · equal coordinate error can mean unequal overlap

Two predictions move one boundary by 10 pixels.

Large object: 200×200

a 10-pixel error is 5% of its width.

Small object: 20×20

the same error is 50% of its width.

SHOULD · equal coordinate error can mean unequal overlap

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Coordinate losses see four residuals. IoU sees the region those residuals create.

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Coordinate losses see four residuals. IoU sees the region those residuals create.

Use coordinate losses for a stable regression signal; report overlap-aware metrics for the geometry the task actually values.

SMOOTH $L_1 \cdot \beta = 1$

quadratic near 0; linear for large residuals

$0.2 \rightarrow 0.02$; $3 \rightarrow 2.5$

GIoU

$$\text{GIoU} = \text{IoU} - \frac{\frac{|C|}{(A \cup B)}}{|C|}$$

C : smallest enclosing box

SHOULD · Smooth L_1 and GIoU repair different failures

D DERIVATION

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Plain 1 – IoU is flat while boxes are disjoint. GIoU changes with the enclosing gap; Smooth L_1 instead controls coordinate-residual robustness.

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Rezatofighi et al. · Generalized IoU

**TRAIN: dense candidates need
selective supervision**

Dense heads create a foreground–background imbalance

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



Dense heads create a foreground–background imbalance

TRAIN candidate $\leftrightarrow$ truth; losses

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If thousands of easy background candidates contribute ordinary cross-entropy, their count can drown out the rare positives and hard negatives.

Focal loss turns confidence into a gradient-volume control

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

$$\text{FL}(p_t) = -\alpha_{t(1-p_t)}^{\gamma} \log p_t$$

Focal loss turns confidence into a gradient-volume control

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$$\text{FL}(p_t) = -\alpha_t^\gamma (1-p_t) \log p_t$$

p_t

probability assigned to
the true binary label

γ

how strongly easy
examples are down-
weighted

α_t

optional class-balance
weight

Focal loss turns confidence into a gradient-volume control

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how strongly easy
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| The next numeric sets $\alpha_t = 1$ and $\gamma = 2$ to isolate the focusing factor.

With $\gamma = 2$, the hard example is about $978 \times$ louder

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

Easy: $p_t = .9$

$$\text{CE} = -\log .9 \approx .10536$$

$$(1 - p_t)^2 = .01$$

$$\text{FL} \approx .0010536$$

Hard: $p_t = .2$

$$\text{CE} = -\log .2 \approx 1.60944$$

$$(1 - p_t)^2 = .64$$

$$\text{FL} \approx 1.03004$$

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$$\frac{1.03004}{.0010536} \approx 977.6 \approx 978 \times$$

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$$\text{FL} \approx 1.03004$$

$$\frac{1.03004}{.0010536} \approx 977.6 \approx 978 \times$$

Focal loss does not delete easy examples; it makes already-correct examples quiet enough that rare mistakes remain audible.

Diagnose training failures as symptom → suspect → test

TRAIN candidate ↔ truth; losses**INFER** prediction ↔ prediction; NMS**EVAL** prediction ↔ truth; AP

symptom	suspect	smallest discriminating test
box loss falls; IoU stays near 0	box format / decode mismatch	round-trip one known anchor and draw decoded boxes
almost no positive anchors	assignment geometry too strict	histogram best IoU per truth; inspect τ_+
objectness predicts background	easy negatives dominate	log pos/neg loss mass; compare focal or sampling

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objectness predicts background	easy negatives dominate	log pos/neg loss mass; compare focal or sampling

Inspect geometry and assignment before changing the backbone: a model cannot learn targets that the pipeline encoded incorrectly.

Follow along: boxes, IoU, matching, and anchor round-trips

I INTERACTIVE

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

INTERACTIVE

↗ colab.research.google.com/github/nipunbatra/dl-teaching/blob/master/notebooks/L09/01_iou_and_bbox_losses.ipynb

Open the exact Colab. Predict the $\frac{1444}{1756}$ IoU, implement the clamp, verify the anchor encode/decode round-trip, and build the held matching ledger before reading its outputs.

Follow along: boxes, IoU, matching, and anchor round-trips

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TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

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BOX

$$[x_0, y_0, x_1, y_1]; \text{no} + 1$$

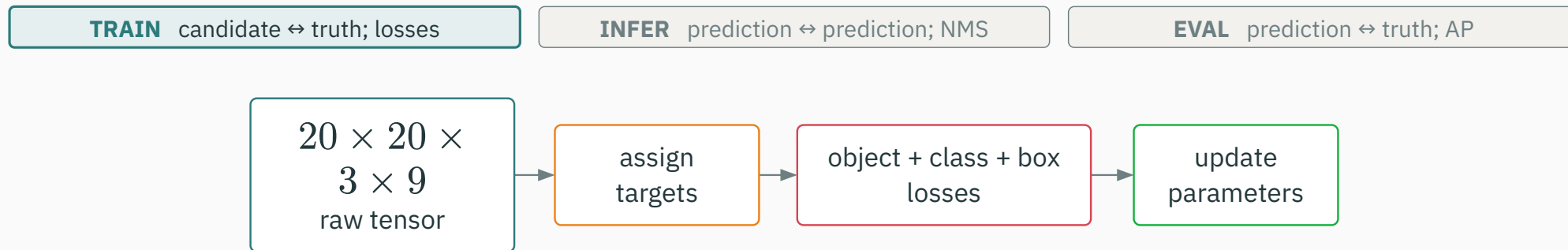
OVERLAP

$$\text{IoU}(A, B) \approx .822$$

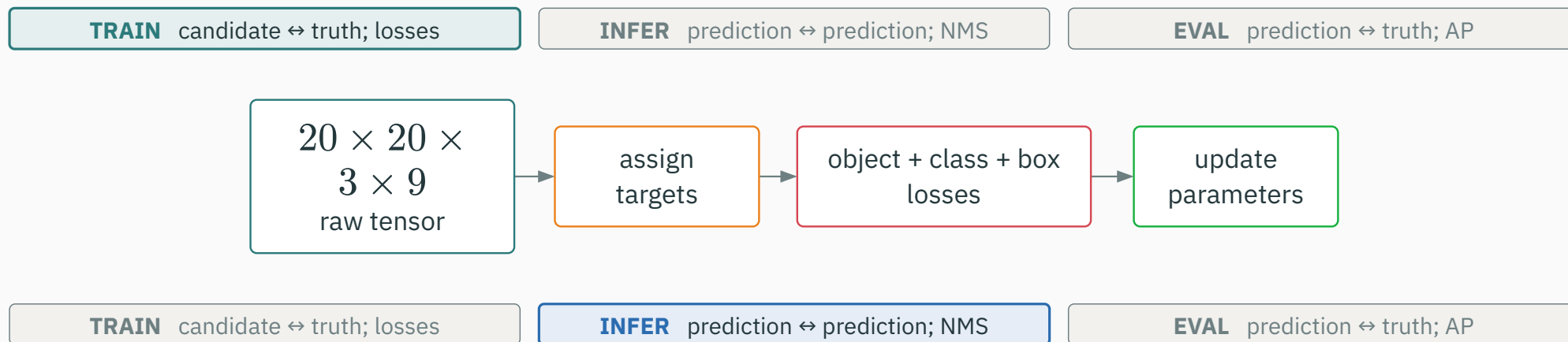
ANCHOR

$$t = (.20, -.10, .693, 0)$$

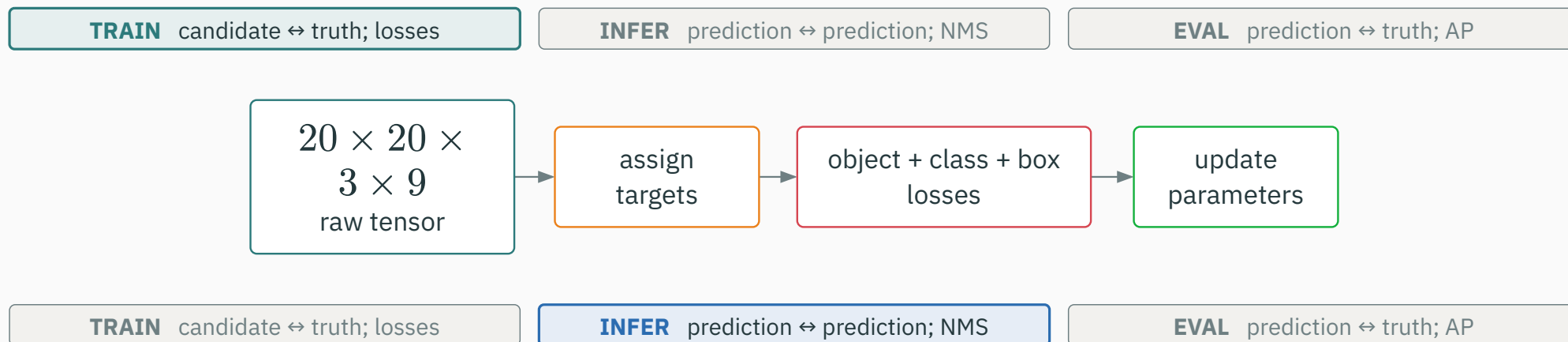
TRAIN closes with targets; inference opens without them



TRAIN closes with targets; inference opens without them



TRAIN closes with targets; inference opens without them



At inference there is no truth, no assignment, and no loss—only decoded boxes, scores, thresholds, and prediction-to-prediction suppression.

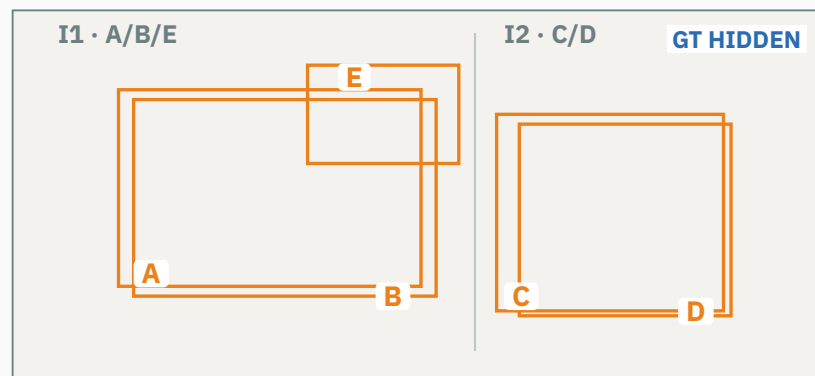
INFER: score first, then suppress duplicates

The constructed raw ledger isolates the inference rule

TRAIN candidate $\leftrightarrow$ truth; losses

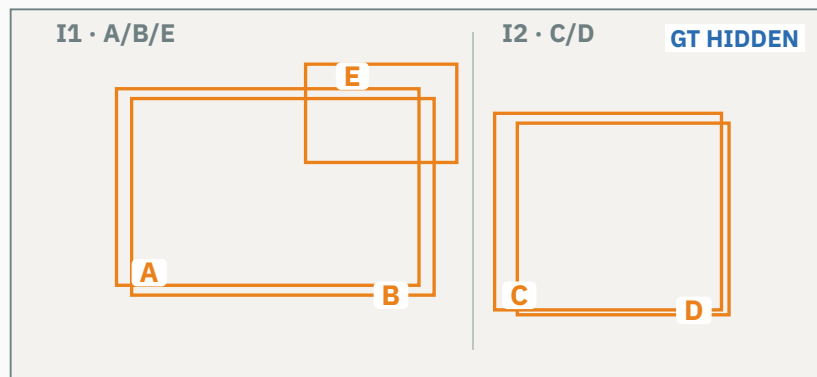
INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



A .95 B .90 E .89 C .88 D .75

The constructed raw ledger isolates the inference rule

TRAIN candidate $\leftrightarrow$ truth; losses**INFER** prediction $\leftrightarrow$ prediction; NMS**EVAL** prediction $\leftrightarrow$ truth; AP

A .95 B .90 E .89 C .88 D .75

A/B/E share (I1, head); C/D share (I2, head). These scores are pedagogical constructions, not model output. INFER sees neither XML truth nor TP/FP labels.

Score thresholding removes weak evidence—not duplicates

QUESTION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

Use a deliberately permissive score threshold $s \geq .70$.

A .95

B .90

E .89

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D .75

All five survive.

Score thresholding removes weak evidence—not duplicates

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EVAL prediction $\leftrightarrow$ truth; AP

Use a deliberately permissive score threshold $s \geq .70$.

A .95

B .90

E .89

C .88

D .75

All five survive.

Thresholding can drop low scores, but it cannot tell that two high-scoring boxes describe the same object.

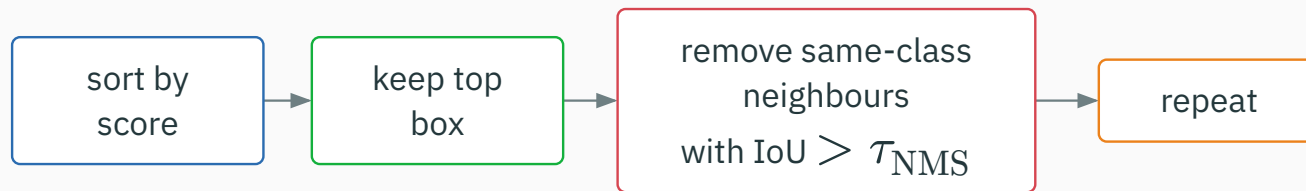
NMS is a greedy prediction-to-prediction comparison

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



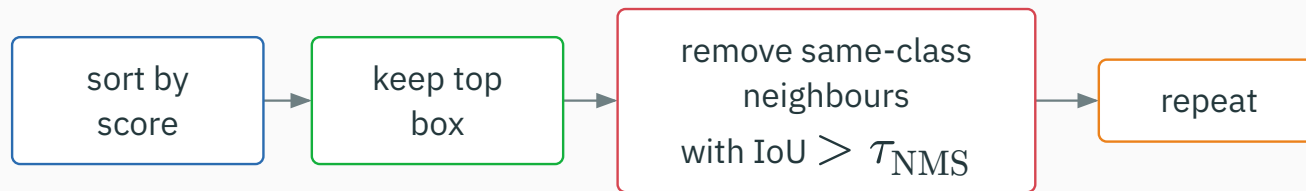
NMS is a greedy prediction-to-prediction comparison

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

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EVAL prediction $\leftrightarrow$ truth; AP



No ground truth participates. Standard NMS has no learned parameter and no gradient; here $\tau_{\text{NMS}} = .50$.

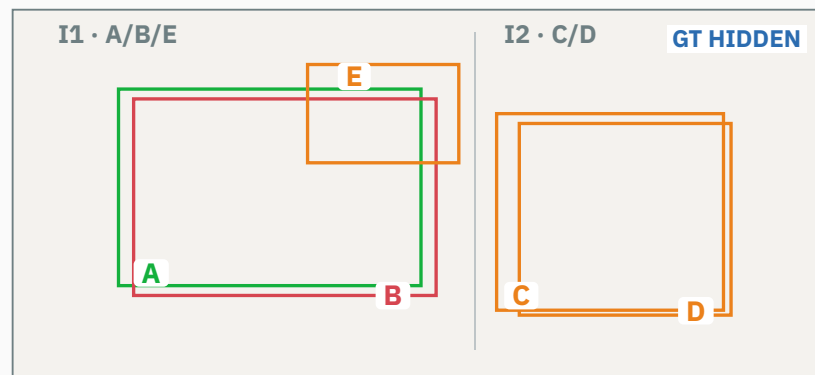
Step 1: A claims its high-overlap neighbourhood

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



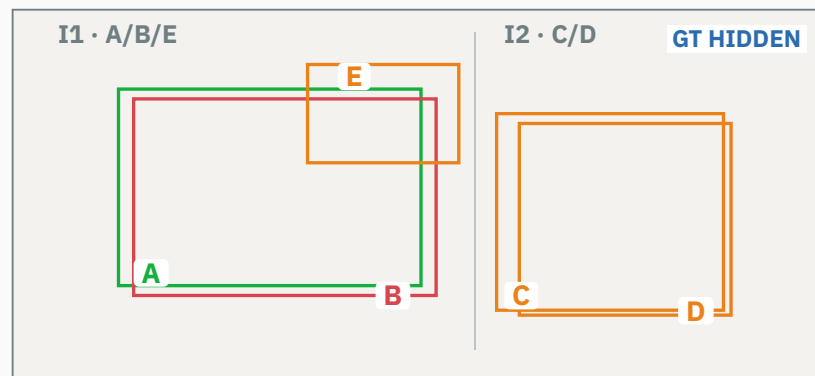
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EVAL prediction $\leftrightarrow$ truth; AP



$$\text{IoU}(A, B) = .822 > .50 \rightarrow \text{keep A, suppress B}$$

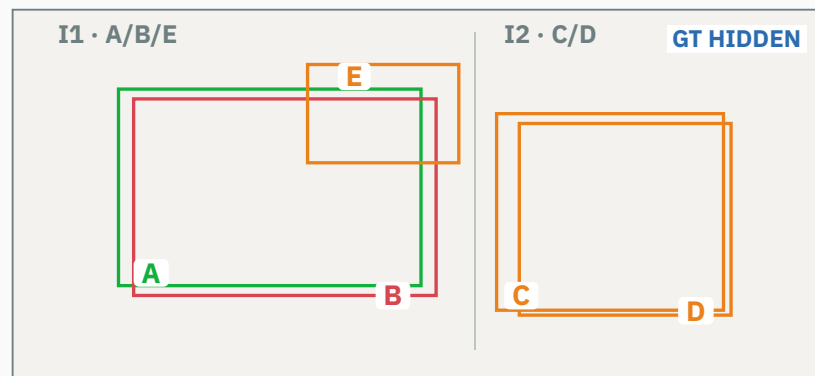
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INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



$$\text{IoU}(A, B) = .822 > .50 \rightarrow \text{keep A, suppress B}$$

E, C, and D remain because none overlaps A above the NMS threshold.

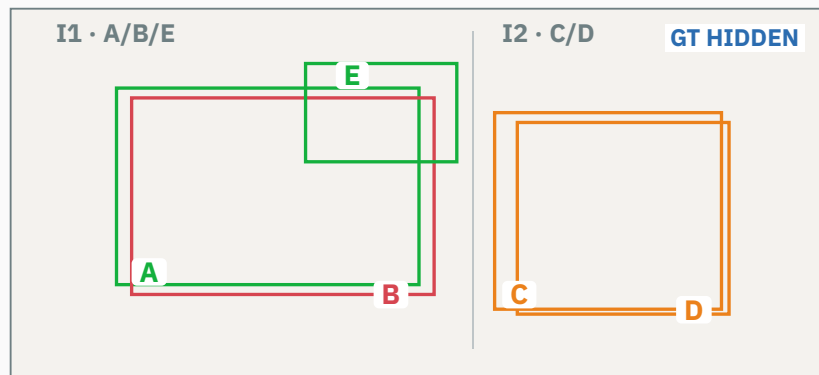
Step 2: E survives because NMS cannot test correctness

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



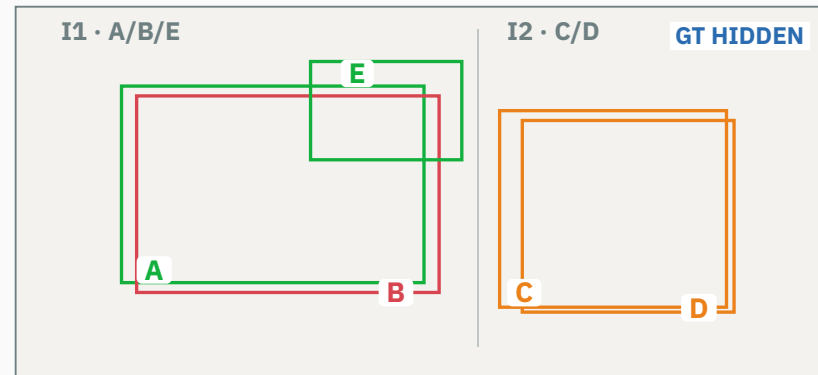
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$E = .89$ is now the highest remaining score; its IoU with every survivor is below .50.

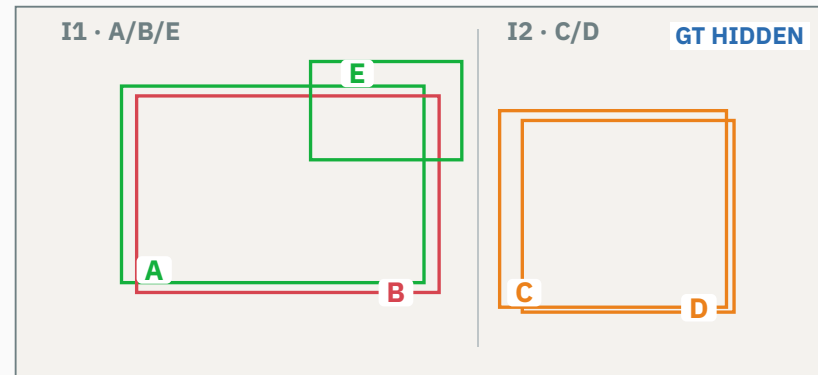
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EVAL prediction $\leftrightarrow$ truth; AP



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Keep E. NMS removes duplicates around a stronger prediction; it does not remove isolated false alarms.

Step 3: C claims the I2 head neighbourhood

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

HELD CASE · LAST PAIR $C = [60, 15, 90, 55]$ and $D = [63, 17, 91, 56]$.

$$A_I = (90 - 63)(55 - 17) = 27 \cdot 38 = 1,026$$

Step 3: C claims the I2 head neighbourhood

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

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EVAL prediction $\leftrightarrow$ truth; AP

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$$A_I = (90 - 63)(55 - 17) = 27 \cdot 38 = 1,026 \quad A_U = 1,200 + 1,092 - 1,026 = 1,266$$

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$$\text{IoU}(C, D) = \frac{1026}{1266} \approx .810 > .50$$

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Keep C; suppress D. No candidates remain.

NMS returns A, E, C—not “the correct boxes”

A ANSWER

TRAIN candidate ↔ truth; losses

INFER prediction ↔ prediction; NMS

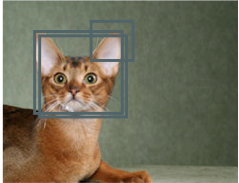
EVAL prediction ↔ truth; AP

NMS RUNS WITHIN EACH IMAGE

Same class · score order · IoU > 0.50 · ground truth never participates

I1 · RAW A/B/E

Abyssinian_1



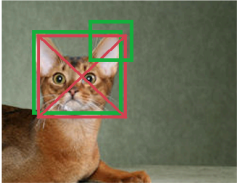
KEEP —

SUPPRESS —

PENDING A, B, E

I1 · RETURN A/E

Abyssinian_1



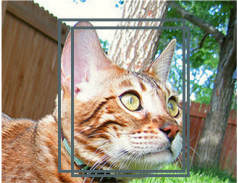
KEEP A, E

SUPPRESS B

PENDING —

I2 · RAW C/D

Bengal_10



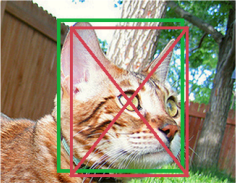
KEEP —

SUPPRESS —

PENDING C, D

I2 · RETURN C

Bengal_10



KEEP C

SUPPRESS D

PENDING —

$\text{IoU}(A,B) = 1444/1756 = 0.822$

$\text{IoU}(C,D) = 1026/1266 = 0.810$

BATCH → A, E, C

No cross-image suppression: A can never suppress C. E survives because it is isolated, not because it is correct.

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TRAIN candidate ↔ truth; losses

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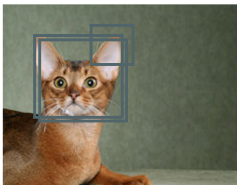
EVAL prediction ↔ truth; AP

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Same class · score order · IoU > 0.50 · ground truth never participates

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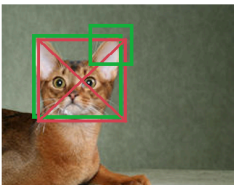
Abyssinian_1



KEEP —
SUPPRESS —
PENDING A, B, E

I1 · RETURN A/E

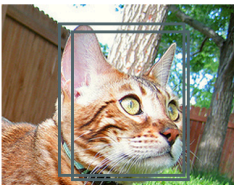
Abyssinian_1



KEEP A, E
SUPPRESS B
PENDING —

I2 · RAW C/D

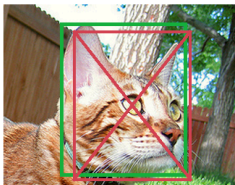
Bengal_10



KEEP —
SUPPRESS —
PENDING C, D

I2 · RETURN C

Bengal_10



KEEP C
SUPPRESS D
PENDING —

$\text{IoU}(A, B) = 1444/1756 = 0.822$

$\text{IoU}(C, D) = 1026/1266 = 0.810$

BATCH → A, E, C

No cross-image suppression: A can never suppress C. E survives because it is isolated, not because it is correct.

NMS is grouped by (image_id, class): A can never suppress C across images. The global returned score order is A, E, C; correctness waits for EVAL.

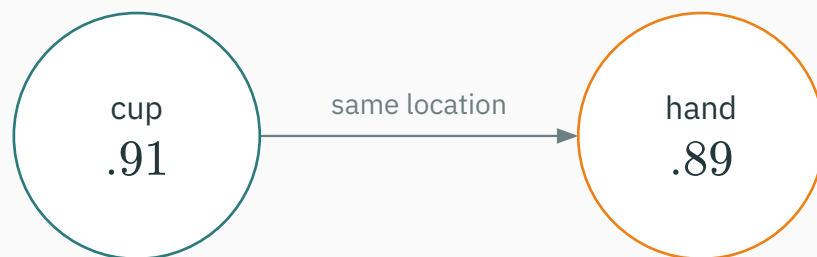
Class-aware NMS prevents a cup from deleting a hand

QUESTION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



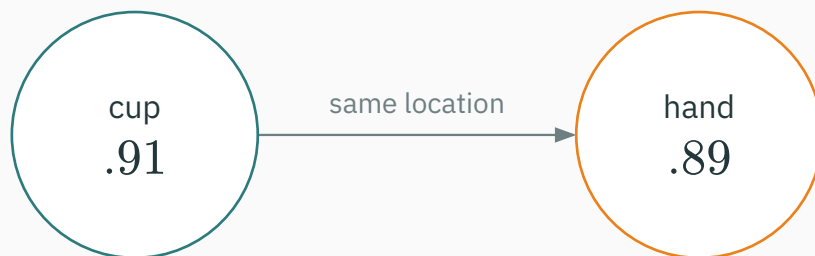
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Usually compare boxes within the same class. Cross-class overlap may describe two real, interacting objects.

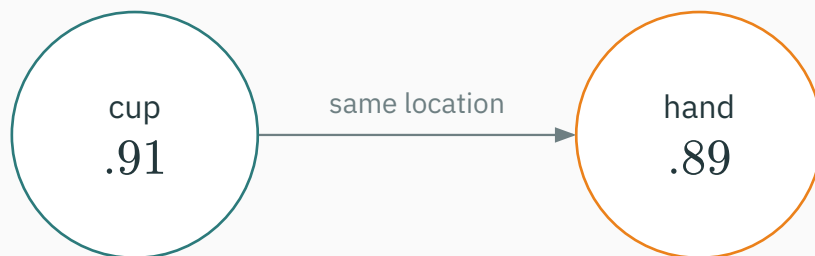
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Usually compare boxes within the same class. Cross-class overlap may describe two real, interacting objects.

A low NMS threshold can still merge two nearby objects of the **same** class. Increasing τ_{NMS} often keeps more boxes for a fixed score order, but keep-count is not a global monotonic law across changed scores/classes/pipelines.

Follow along: print every NMS decision, then break it

I INTERACTIVE

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

INTERACTIVE

↗ colab.research.google.com/github/nipunbatra/dl-teaching/blob/master/notebooks/L09/02_nms_from_scratch.ipynb

Open the exact Colab. Predict A, E, C; print the keep/suppress trace; verify class-aware NMS; then construct two distinct same-class objects that a low threshold wrongly merges.

Follow along: print every NMS decision, then break it

I INTERACTIVE

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Open the exact Colab. Predict A, E, C; print the keep/suppress trace; verify class-aware NMS; then construct two distinct same-class objects that a low threshold wrongly merges.

KEEP

A .95, E .89, C .88

SUPPRESS

B by A; D by C

LIMIT

isolated false alarm E survives

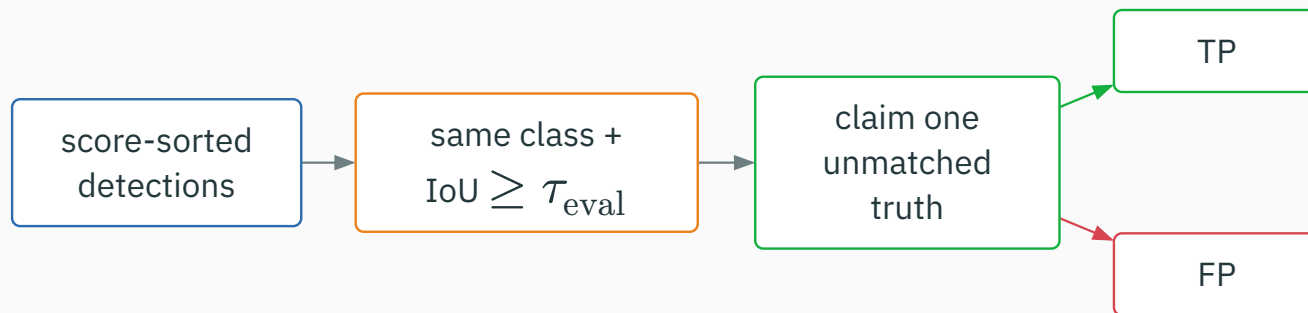
EVAL: match the returned set to truth, one claim at a time

Evaluation matching is not training assignment

TRAIN candidate $\leftrightarrow$ truth; losses

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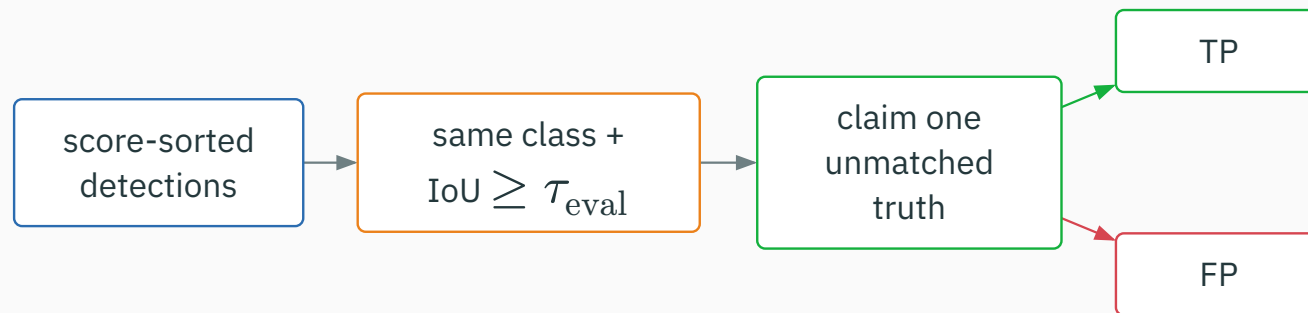


Evaluation matching is not training assignment

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At $\tau_{\text{eval}} = .50$, each truth may be claimed once. A duplicate would be an FP even if its IoU were high.

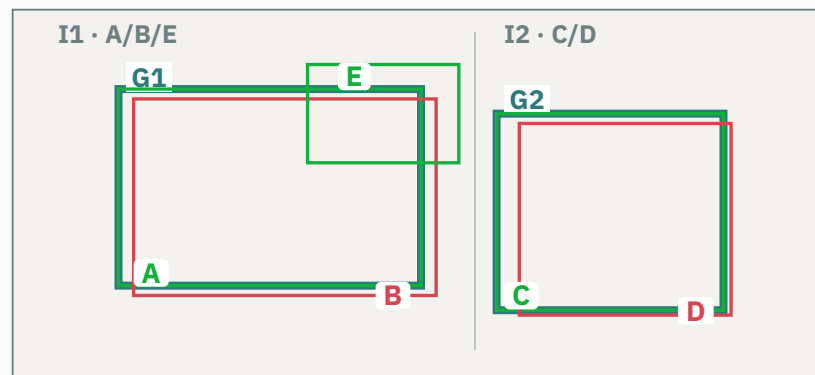
Commit: which of A, E, C are true positives?

QUESTION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP



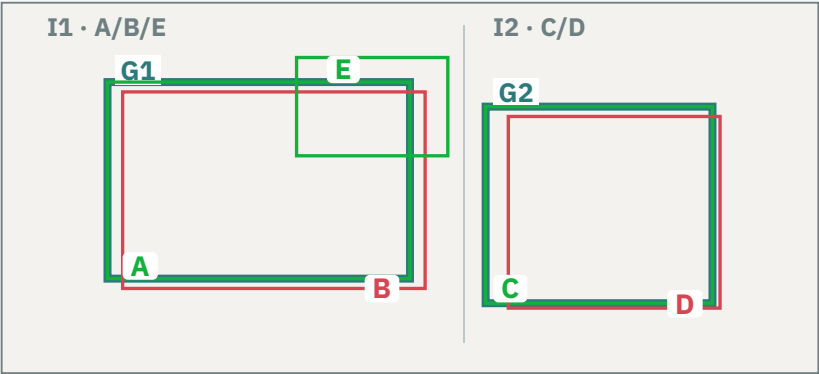
Commit: which of A, E, C are true positives?

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detection	score	best same-class IoU	commit
A	.95	1.000 with G_1	?
E	.89	.127 with G_1	?
C	.88	1.000 with G_2	?

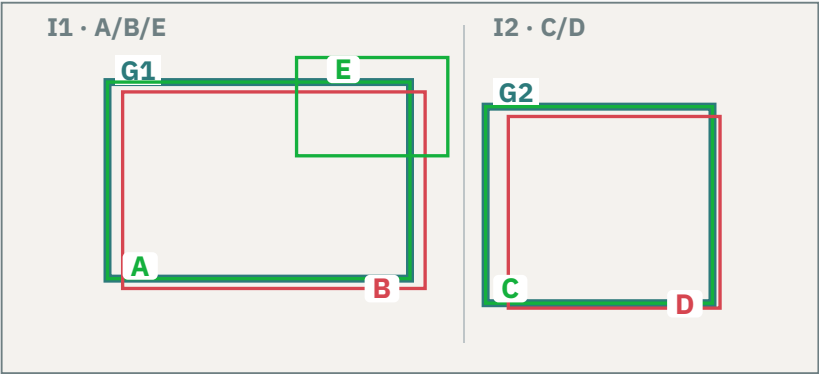
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Match only within image_id, then concatenate survivors into global score order. A truth claimed by an earlier detection is unavailable to later detections in that image.

Evaluation exposes the false alarm NMS could not see

A ANSWER

TRAIN candidate ↔ truth; losses

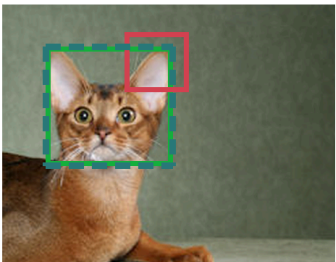
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EVAL prediction ↔ truth; AP

MATCH THE RETURNED BATCH TO BOTH REAL TRUTHS

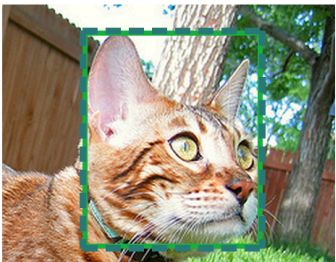
Global score order A, E, C · one claim per XML box

I1 · G1



A = TP · E = FP

I2 · G2



C = TP

E overlaps G1 by only $9/71 \approx 0.127$; NMS could not know it was wrong.

Evidence taxonomy: photos/XML = observed · candidates/scores/crops = constructed · IoU/NMS/matching/AP = computed · model output = none.

RANKED MATCHING

RANK	BOX	IOU	RESULT
1	A 0.95	1.000	TP
2	E 0.89	0.127	FP
3	C 0.88	1.000	TP

AVERAGE PRECISION

$$\text{AP} = (1 + 2/3) / 2 \\ = 5/6 \approx 0.833$$

FINAL COUNTS

2 TP · 1 FP · 0 FN

Two-image teaching batch; not a benchmark.

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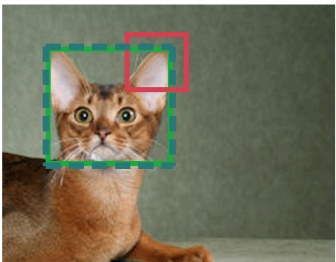
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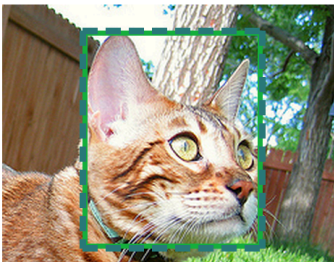
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GT is revealed only now: A claims I1/G1, E is an I1 false alarm at $\frac{9}{71}$, and C claims I2/G2. Two TPs, one FP, zero FNs.

Precision and recall evolve with the score ranking

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

after	cum. TP	cum. FP	precision	recall
A	1	0	$\frac{1}{1} = 1.000$	$\frac{1}{2} = .500$

Precision and recall evolve with the score ranking

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after	cum. TP	cum. FP	precision	recall
A	1	0	$\frac{1}{1} = 1.000$	$\frac{1}{2} = .500$
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A	1	0	$\frac{1}{1} = 1.000$	$\frac{1}{2} = .500$
E	1	1	$\frac{1}{2} = .500$	$\frac{1}{2} = .500$
C	2	1	$\frac{2}{3} = .667$	$\frac{2}{2} = 1.000$

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Lowering the score threshold admitted E before C: precision fell, then recall rose when C arrived.

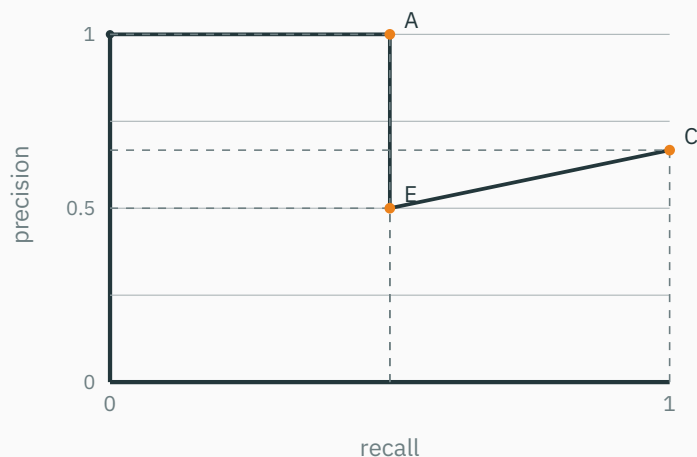
The toy AP convention averages precision at TP ranks

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At the two TP ranks:

$$p_A = 1$$

$$p_C = \frac{2}{3}$$

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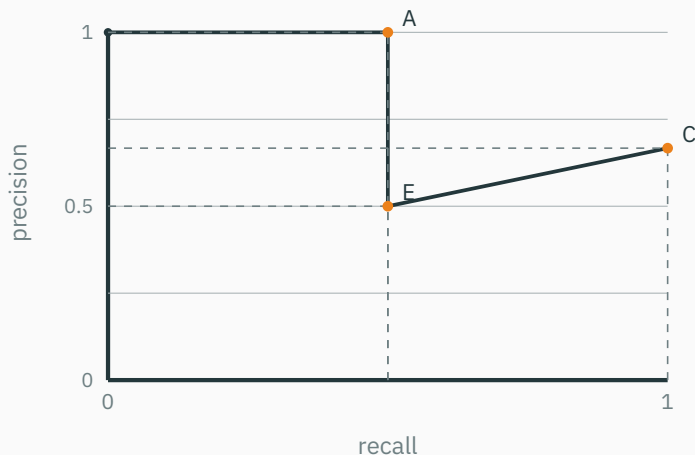
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After NMS: A, E, C; B and D were suppressed before evaluation. Axes run from 0 to 1.

This TP-rank average is a transparent teaching convention. It is not a claim about every benchmark's interpolation rule.

SHOULD · COCO AP fixes the full evaluation protocol

D DERIVATION

TRAIN candidate $\leftrightarrow$ truth; losses

INFER prediction $\leftrightarrow$ prediction; NMS

EVAL prediction $\leftrightarrow$ truth; AP

LOCALIZATION

10 IoU thresholds .50, .55, ..., .95

INTERPOLATION

101 recall thresholds 0, .01, ..., 1

AGGREGATION

average over classes; declare area +
max detections

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Say AP50 for one IoU cutoff. “COCO AP” normally means the average across IoU thresholds; the official API also declares area ranges and maxDets.

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Official COCO evaluation implementation

Diagnose metric surprises on the correct clock

TRAIN candidate $\leftrightarrow$ truth; losses**INFER** prediction $\leftrightarrow$ prediction; NMS**EVAL** prediction $\leftrightarrow$ truth; AP

symptom	suspect	smallest discriminating test
high scores; low AP	ranking is confidently wrong	inspect first FP for class, IoU, duplicate status
AP50 high; COCO AP low	boxes are loose	compare AP50 vs AP75; histogram matched IoU
recall caps below 1	thresholding / missing candidates	lower score threshold before NMS; count unmatched truths

TRAIN candidate ↔ truth; losses

INFER prediction ↔ prediction; NMS

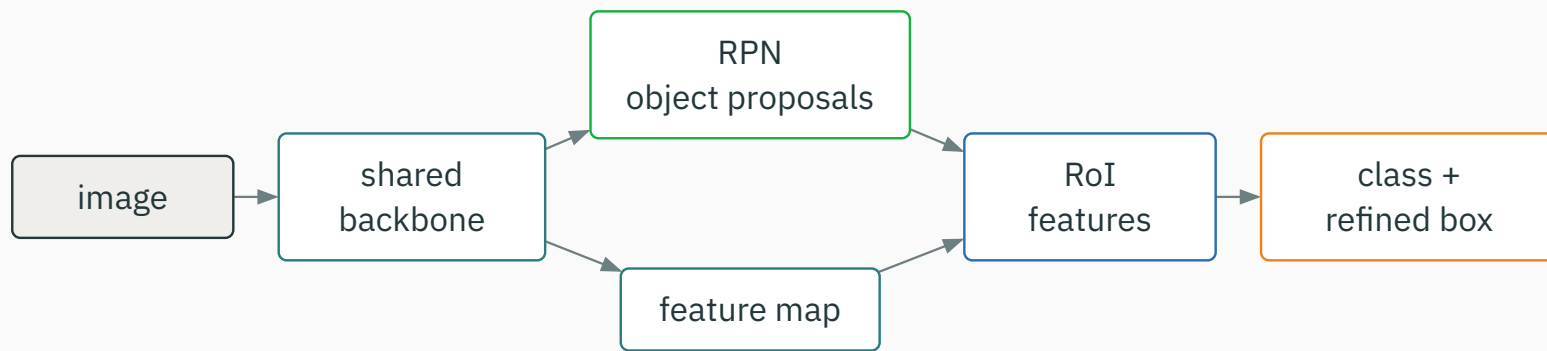
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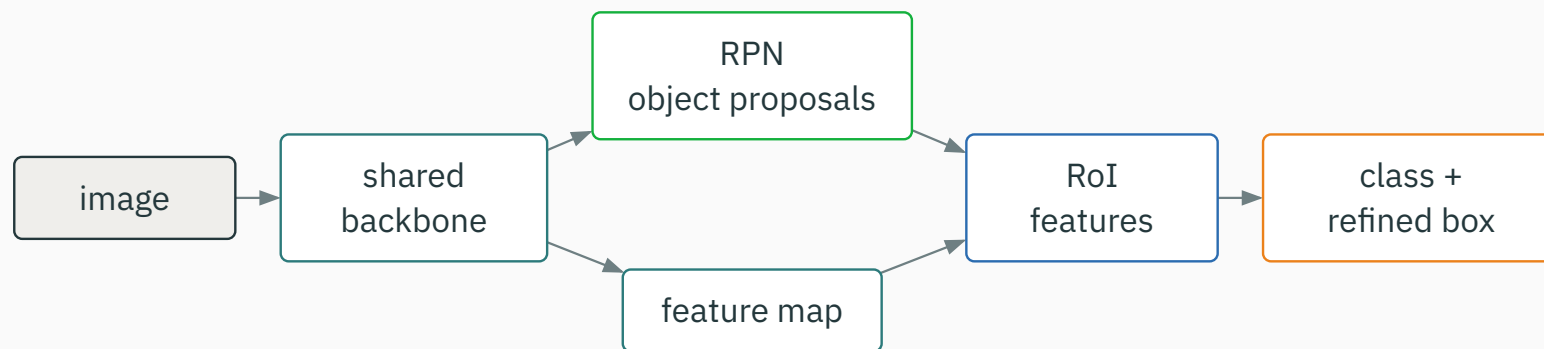
Loss, NMS, and AP expose different failure surfaces. Name the clock before changing a threshold.

**SHOULD: architectures rearrange the same
evidence**

SHOULD · two-stage means propose, then refine



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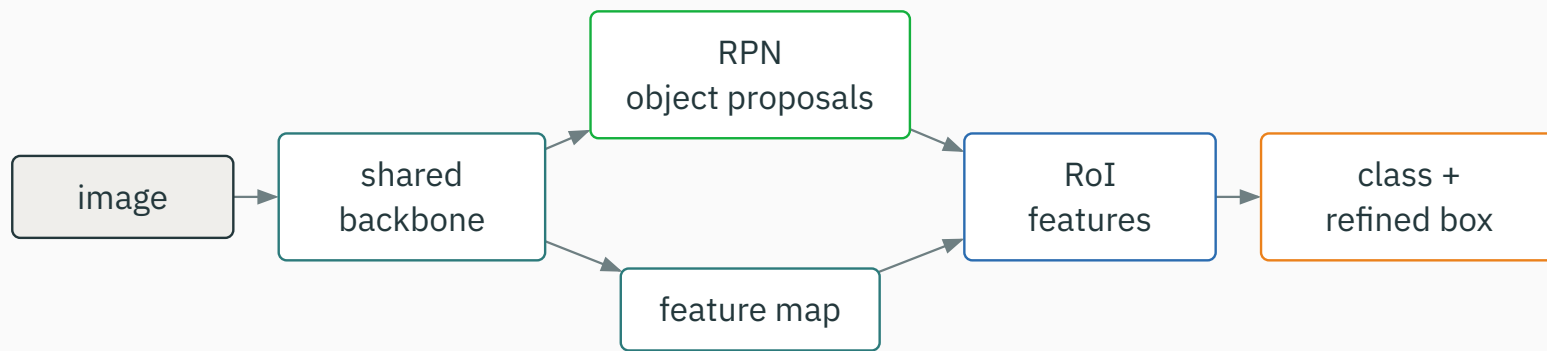
RPN

Region Proposal Network: predicts objectness + proposal boxes densely.

ROI

Region of Interest: samples a fixed-size feature for each proposal.

SHOULD · two-stage means propose, then refine



RPN

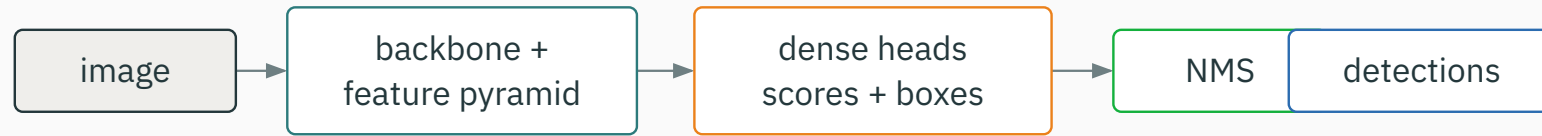
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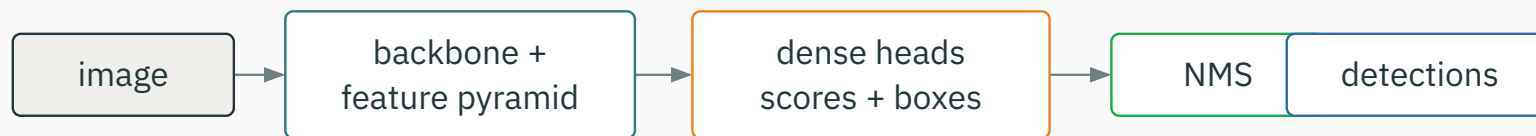
Region of Interest: samples a fixed-size feature for each proposal.

Stage 1 asks “where might an object be?” Stage 2 spends more computation to classify and refine a smaller proposal set.

SHOULD · one-stage predicts every candidate directly

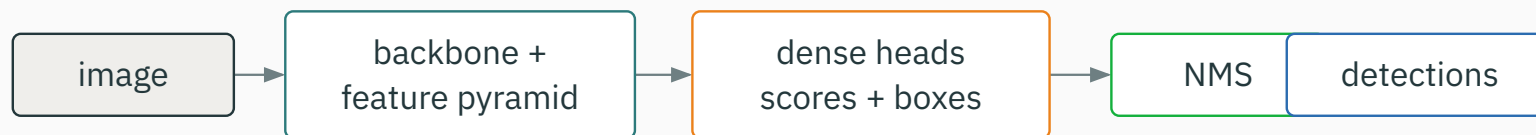


SHOULD · one-stage predicts every candidate directly



There is no learned proposal stage or per-RoI second head: classification and localization happen densely in one pass.

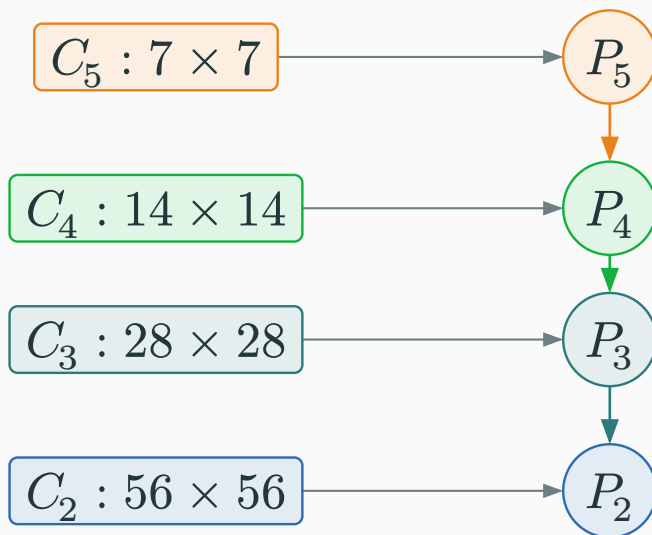
SHOULD · one-stage predicts every candidate directly



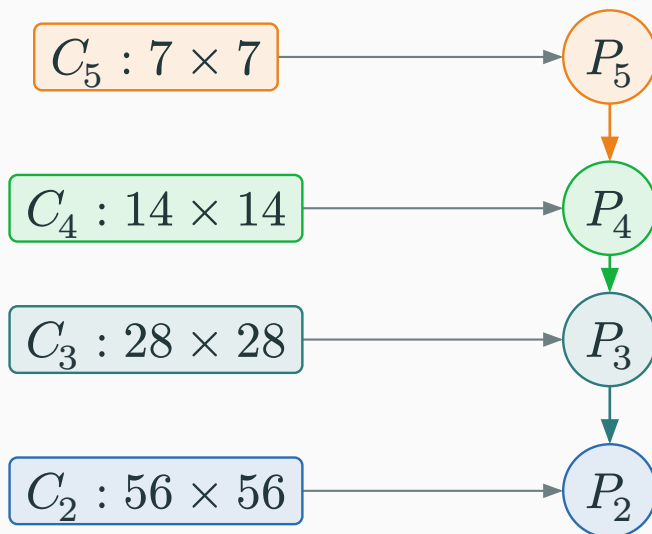
There is no learned proposal stage or per-RoI second head: classification and localization happen densely in one pass.

One-stage versus two-stage describes where proposal/refinement computation occurs—not a timeless guarantee about speed or accuracy.

SHOULD · FPN gives small and large objects strong features

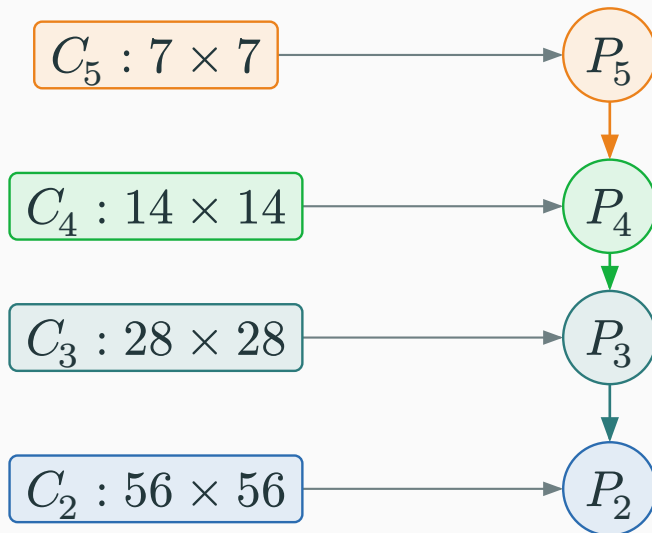


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FPN = **Feature Pyramid Network**: lateral 1×1 projections + top-down upsampling + same-shape addition.

SHOULD · FPN gives small and large objects strong features



FPN = **Feature Pyramid Network**: lateral 1×1 projections + top-down upsampling + same-shape addition.

High-resolution levels localize small objects; top-down semantics make those levels useful for recognition.

ANCHOR-BASED

start from (x_a, y_a, w_a, h_a) ; regress normalized translations and log scales

ANCHOR-FREE

predict a center/keypoint or distances to left, top, right, bottom sides

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ANCHOR-FREE

predict a center/keypoint or distances to left, top, right, bottom sides

Both still need spatial features, class evidence, box geometry, a training matcher, and an evaluation protocol.

SOFT-NMS

decay a neighbour's score instead of deleting it; nearby objects may survive

SET PREDICTION

train a fixed query set with one-to-one assignment;
duplicate avoidance moves into learning

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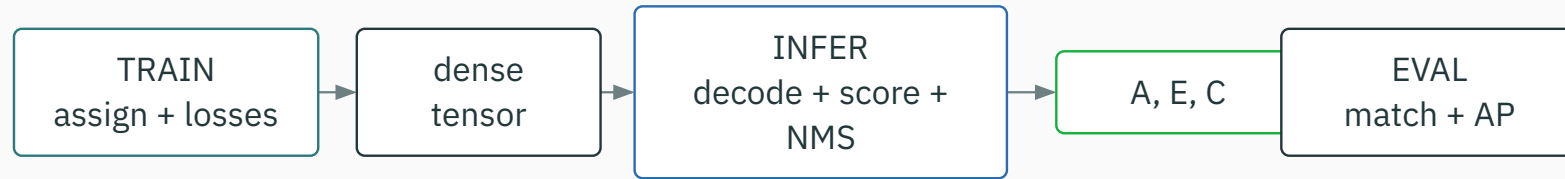
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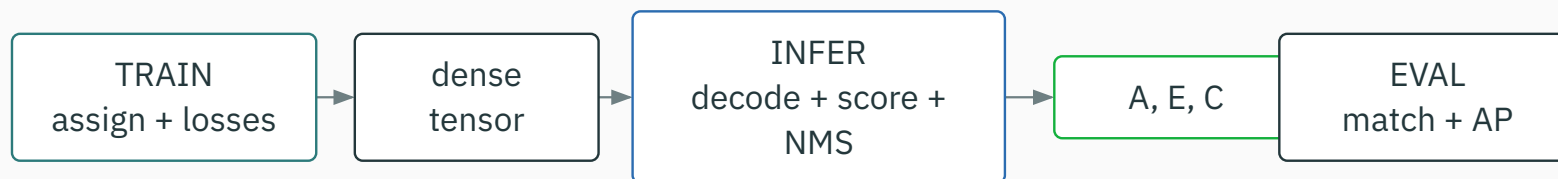
These alternatives change inference and training, but the evaluation clock still matches scored predictions to truths under a declared protocol.

One pipeline, three clocks, one returned set

The complete pipeline changes what IoU is comparing

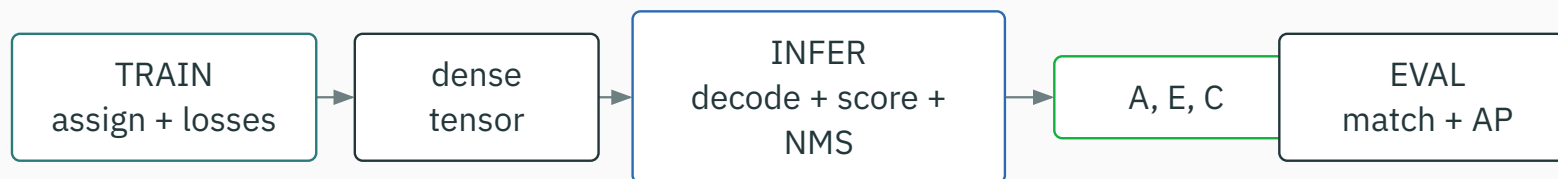


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clock	IoU compares	decision
TRAIN	candidate ↔ truth	which targets and losses apply
INFER	prediction ↔ prediction	which neighbours are redundant
EVAL	prediction ↔ truth	TP / FP / FN and AP

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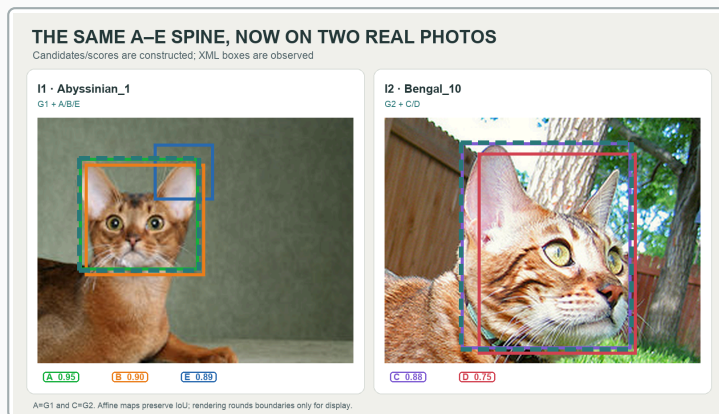
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The formula is the same; the objects, threshold, and consequence are not.

Revisit the opening commitment before revealing it

QUESTION

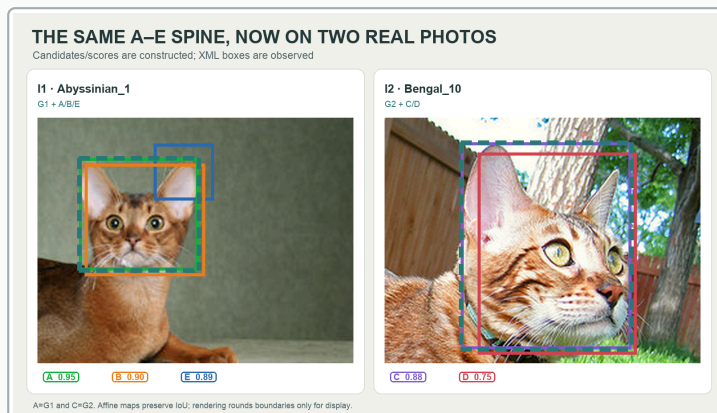
HELD CASE · SAME FIVE CANDIDATES A .95, B .90, E .89, C .88, D .75; $\tau_{\text{NMS}} = .50$, $\tau_{\text{eval}} = .50$.



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A · THRESHOLD

all five pass .70

B · GEOMETRY

decode → score → NMS → evaluation
match

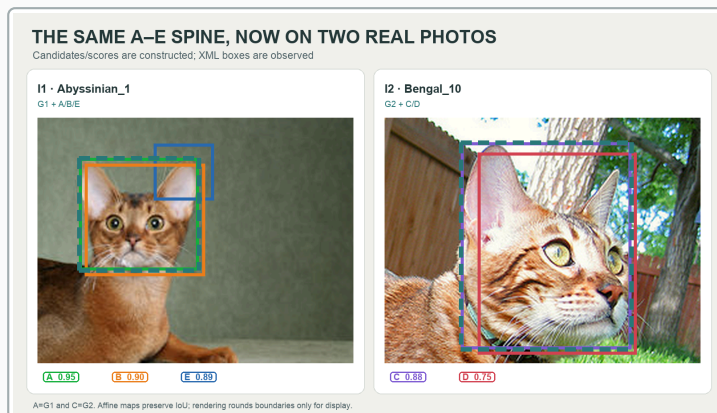
C · LOSS

requires truths unavailable at
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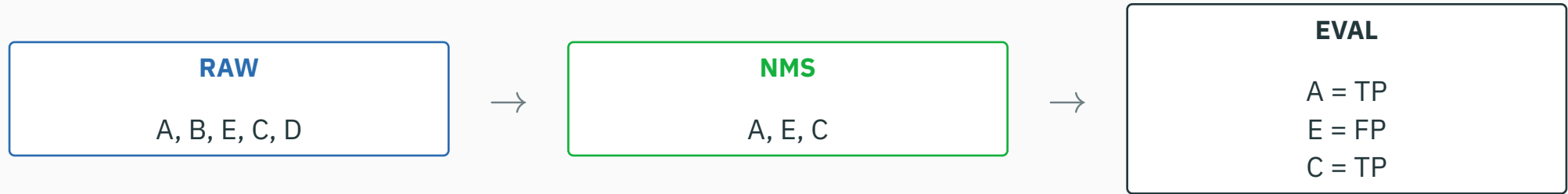
C · LOSS

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Defend your original choice using the exact boxes and clocks—not a detector name.

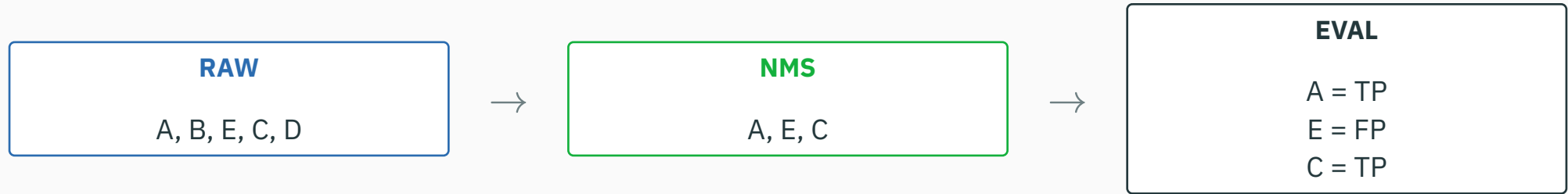
Answer: B is complete—and still cannot remove E

A ANSWER



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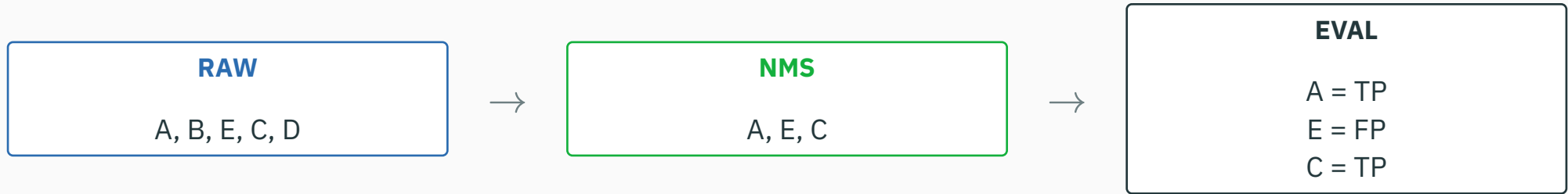
A ANSWER



$$AP_{\text{toy}} = \frac{5}{6} \approx .833$$

Answer: B is complete—and still cannot remove E

A ANSWER



$$AP_{\text{toy}} = \frac{5}{6} \approx .833$$

Thresholds control admission, NMS controls redundancy, and matching measures correctness. No one operation can substitute for the others.

Given a new detector, ask in order:

1. What box coordinate and area convention does every component use?
2. What is the dense output shape, and what does each scalar mean?
3. Which TRAIN matcher creates positive, ignored, and background targets?
4. Which loss reduction and weights determine gradient mass?
5. Which INFER score and NMS policies create the returned set?
6. Which EVAL IoU, interpolation, class, area, and max-detection protocol defines AP?
7. Which symptom → suspect → test check will distinguish geometry from learning failure?

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A detector is trustworthy only when the tensor, geometry, three clocks, and metric protocol are all explicit.

L11 handoff: boxes become a prediction at every pixel

DETECTION

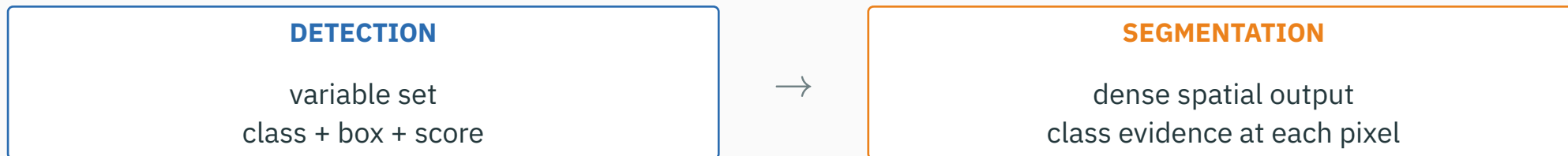
variable set
class + box + score



SEGMENTATION

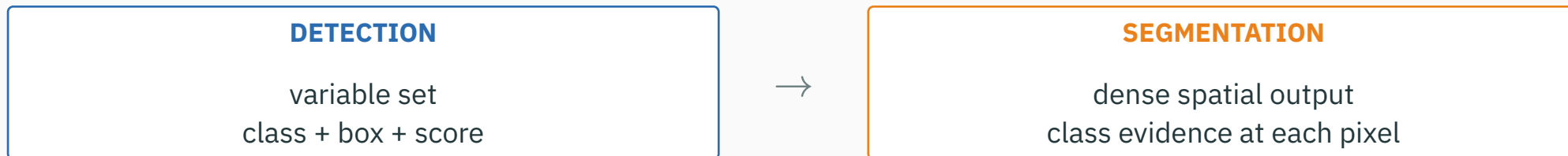
dense spatial output
class evidence at each pixel

L11 handoff: boxes become a prediction at every pixel



Both reuse the L8B backbone. L11 will replace four-number rectangles with masks, then recover boundaries that pooling made coarse.

L11 handoff: boxes become a prediction at every pixel



Both reuse the L8B backbone. L11 will replace four-number rectangles with masks, then recover boundaries that pooling made coarse.

Today asked “which objects, and where?” Next asks “which class owns every pixel—and which instance?”

DETECTOR FAMILIES

Girshick et al. · R-CNN
Girshick · Fast R-CNN
Ren et al. · Faster R-CNN
Redmon et al. · YOLO

LOSSES, PYRAMIDS + CODE

Lin et al. · Focal Loss / RetinaNet
Lin et al. · Feature Pyramid Networks
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REAL EVIDENCE

Oxford-IIIT Pet · Parkhi et al. (CVPR 2012)
I1 Abyssinian_1 · I2 Bengal_10
official XML tight-head ROIs · CC BY-SA 4.0

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shared/vision-evidence/oxford-iiit-pet/19/
builder + manifest + CSV decision ledgers
A–E/scores = constructed · model output = none

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